Justification and Decisions

HEL-8048

1 Figures

The figures used in this project is a combination of what is found in weather analysis and to help explain later decisions. The aim was also to make it as readable as possible, even when presenting a lot of information at one time. The use of heatmaps is an attempt to show daily temperature data over almost a century and to make vertain trends visible and are much more successful than the linegraph before. Said linegraph is also included to show how easy it can be to create figures which are difficult to read and interpret. The use of geographical figures is to show where we collect data and the inclusion of height lines (figure 1) makes it easier to understand the geographical nature around each station.

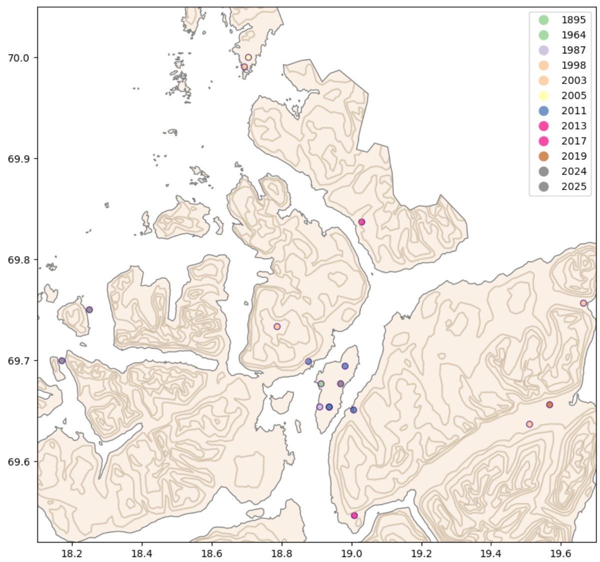

Figure 1: Geographical figure with all weather staions

Colour choices were made to try and make the graphs as readable as possible and to highlight certain information, as well as following established rules found in these types of analysis. The Annual Average Temperature Anomaly graphs, shows positive temperatures in red and negative in blue with a gradient going from each point. While a single font was not used for all the graphs in this project, this was done as some fonts were difficult to read in different graphs. It was also not deemed worthy to invest enough time to find a unifying font that would work for all the graphs, instead finding the ones that work on an individual level worked best. When directly comparing colours to each other, gradients were used instead to try to make things more friendly to read, such as in the heatmaps, or make the colours highlight the differences, but not be the only indicator as in the boxplot used.

2 Discussion

The notebook for the main project tries to justify and explain the different plots using markdown cells. This was also done to help make the code easier to read, as there isn't as many comments in the code itself. The use of more descriptive variable names and markdown cells were tried to use instead of comments as they can be seen as disrupting when reading the code for some.

This project uses a variety of statistical techniques to analyse the data, with averages, normals, distributions and other methods. In combination with markdown cells and graphs the goal is for the reader to gain a certain understanding in how the weather in Tromsø has changed over time.

To make sure this data is reproducible, environment files with packages and their versions have been included and the FROST API is available to the public, where anyone can repeat the exact procedure as done in this project.